ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE E · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice e.

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How to mark: each ✓ is one mark. Accept other correct answers and wording. Circuits may be drawn in different but correct ways: mark each required element and correct connection.

QUESTION 1

1.1 A semiconductor device with two terminals ✓ that allows current to flow in one direction only (forward bias) and blocks it in the other ✓. (2)

1.2 Anode ✓, cathode ✓, gate ✓. With the anode positive to the cathode (forward biased) ✓, a small current pulse into the gate switches the SCR on, and it stays on until the anode current drops below the holding current ✓. (5)

1.3 Square wave drawn with vertical edges and flat tops ✓; high level ✓ and low level ✓ labelled; amplitude (0 to peak) ✓; period (one full cycle) ✓ (5)

1.4 Any TWO, ✓✓ each: a microcontroller has the CPU, memory (RAM and flash) and input/output ports on one chip, so few extra parts are needed; it is small, cheap and uses little power (good for battery robots); it has built-in ADC, PWM, timers and communication for sensors and motors; it is easy to program for one dedicated task. (4)

1.5 Correct transistor symbol with the emitter arrow pointing INWARDS (towards the base) ✓✓; base ✓; collector and emitter labelled ✓ (4)

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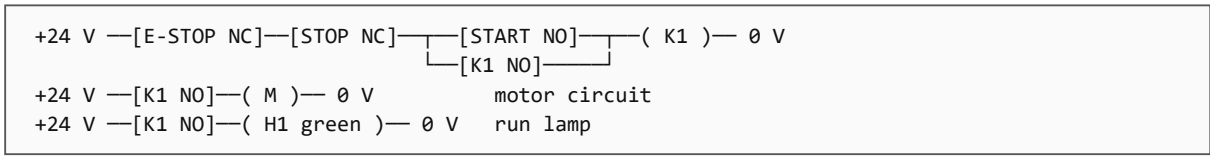
QUESTION 2

2.1 The supply is switched on and off very quickly at a fixed frequency ✓; the width of the ON pulse (duty cycle) is changed ✓; the average voltage across the motor is proportional to the duty cycle ✓, so a wider pulse gives a higher average voltage and a higher speed ✓. (4)

2.2 Controller (e.g. microcontroller / PLC giving step and direction signals) ✓; stepper driver / driver IC ✓; power supply ✓; stepper motor ✓ (also accept indexer, limit switches) (4)

2.3 Open loop: no feedback, the output is not measured or corrected ✓ – e.g. a conveyor on a timer, a stepper without an encoder ✓. Closed loop: the output is measured and fed back to correct errors ✓ – e.g. a servo motor with an encoder, temperature control of an oven ✓. (4)

2.4 Supply lines (+24 V and 0 V) ✓; E-stop (NC) ✓ and STOP (NC) ✓ in series; START (NO) ✓ with K1 holding contact in parallel ✓; relay coil K1 ✓; K1 contact switching the motor ✓; green lamp switched by a K1 contact (or in parallel with the motor) ✓, e.g. (8)



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QUESTION 3

3.1.1 Circle with ONE solid triangle pointing outwards (flow out) ✓, drive shaft / inlet and outlet lines ✓ (2)

3.1.2 Square valve with an arrow offset from the flow line (normally closed) ✓; adjustable spring ✓; dashed pilot line from the inlet ✓ (3)

3.1.3 Cylinder barrel with piston and rod ✓ and TWO ports, one at each end ✓ (2)

3.2 Pressure applied to a confined liquid is transmitted equally in all directions ✓✓. Application ✓: hydraulic press, car hoist / jack, hydraulic brakes, robot or excavator arm. (3)

3.3 Any FOUR ✓✓✓✓: transmits power; lubricates moving parts; seals clearances between parts; carries away heat (cooling); carries contaminants to the filter; protects against rust and corrosion. (4)

3.4 Reservoir, pump (with motor) and filter ✓; pressure relief valve from the pressure line back to tank ✓; 4/3 manually operated valve with tandem centre (P to T in the middle position) ✓; double-acting cylinder with both ports connected to A and B ✓; pilot-operated check valve in the line from the cylinder's lifting (cap) end ✓ with its pilot line from the other cylinder line so it opens to lower ✓ (6)

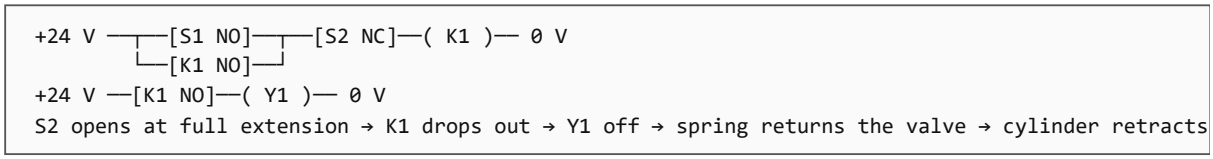
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QUESTION 4

4.1 When the coil is energised, the electromagnet pulls the plunger / armature, which shifts the valve spool ✓ and changes the direction of flow of the oil ✓. (2)

4.2 An NO push button's contacts are open until it is pressed (it closes the circuit when pressed) ✓ – e.g. START ✓. An NC push button's contacts are closed until it is pressed (it opens the circuit when pressed) ✓ – e.g. STOP or emergency stop ✓. (4)

4.3 Working circuit: power pack (pump, relief valve, tank) ✓; 4/2 single-solenoid valve Y1 with spring return ✓; double-acting cylinder correctly connected (P to the cap end when Y1 is on) ✓; limit switch S2 shown at the end of the stroke ✓. Control circuit: +24 V and 0 V lines ✓; S1 (NO) ✓ with K1 holding contact in parallel ✓; S2 (NC) in series with K1 coil ✓; K1 coil ✓; separate rung with a K1 contact switching Y1 ✓, e.g. (10)



4.4 Any FOUR ✓✓✓✓: switch off the pump and lock out / isolate the supply before working; release (bleed) all pressure from the system and accumulators; lower or support suspended loads first; never feel for leaks with your hand (oil jets can penetrate skin); wear safety glasses and gloves; use only parts and hoses rated for the system pressure; clean up spilled oil immediately. (4)

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QUESTION 5

5.1 In order ✓✓✓✓✓: collect information about the fault (talk to the operator, look at the symptoms); study the circuit diagrams and how the system should work; identify the possible causes / isolate the faulty section; test and measure to find the faulty component; repair or replace it and test the whole system; record the fault and the repair. (5)

5.2 Any THREE causes with remedies, ✓✓ each: worn pump (low flow) – repair or replace the pump; leaking piston seals – replace the seals; flow control valve set too low – adjust it; relief valve set too low or stuck open – adjust / repair it; low oil level or blocked suction filter – top up oil / clean or replace the filter; internal leakage in the directional valve – repair or replace the valve. (6)

5.3 Set the meter to the correct DC (or AC) voltage range, higher than the expected voltage ✓; red lead in the V socket and black lead in COM ✓; with the circuit switched on, hold the probes across the two terminals of the coil (in parallel) without touching the metal tips ✓; read the value, then remove the probes and switch the meter off ✓. (4)

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5.4 Hot oil becomes thin (lower viscosity), so leakage increases and lubrication is poorer ✓; seals and hoses are damaged and the oil breaks down (oxidises) faster ✓; components wear quickly and the system loses efficiency ✓. Any TWO causes ✓✓: relief valve opening continuously (set too low); oil level too low; blocked cooler / dirty oil; worn pump with internal leakage; wrong oil viscosity; system working at too high a pressure for too long. (5)

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TOTAL: 100

END OF MARKING GUIDELINE